Group structure

What groups of pilots are and what holds them together

Matteo Di Sante

23 October 2026 • analyses: `xc-lab/analyses/groups` • grouping run `d120_eh500_ez300_th45_g120_q0.1`

Question. What are the groups of the FFVL archive, what holds them together, who leads them and with whom do pilots fly? **Data.** 40 051 configurations in 17 169 episodes, from 38.1-min contacts (identification note); surrogate pairs of its null model as chance.
Findings.

- **Mostly pairs:** 63% of the configurations, 77% of their time. Episodes last 58 min (median); large configurations change members every few minutes and are strings along the route (at 8+ pilots the diameter is 4 or more 61% of the time).
- **Same pace, not a spring:** in contact, the delay of group pilots wanders at 24.5 s^2/s, against 46 for chance contacts as long; the pull back is about the same. After a long first contact they are 1.0 km apart an hour later, against 2.4 km by chance.
- **Sizes from one pilot at a time:** joins and leaves reproduce the size distribution within 4%; the tail is heavy because members of larger groups leave less often ($\gamma = 0.30$).
- **Leadership is fluid:** 3.4 switches per hour; the leader has more experience in 58.2% of the pairs and the higher wing class in 63.9%.
- **Chosen partners:** 37.6% of the pair-days are on repeated pairs (25.7% at random), 32.1% within a club (24.0%).
- **Groups form and break in every phase**, mostly around midday; their shape does not change the flight.

1 Setup

Groups are those of the identification note (`ideas/group-identification`) and of the report *Group and solo flights*, with the same data and run ($\delta = 2$ min, $q = 0.1$, $\tau^* = 38.1$ min). The terms below are graphs; Appendix A defines them from scratch.

- **Pair interval.** Two pilots in contact for at least τ^*: one passes close to where the other was up to δ before or after.
- **Configuration.** At each instant the pair intervals join pilots into a graph; a configuration is one of its connected components, with fixed members. A change of members ends it and starts another.
- **Episode.** A chain of configurations, each starting when the previous one ends and keeping one of its members.
- **How a configuration ends.** Join (members added), leave (members removed in flight), landing (a removed member lands within 5 s, the grid step), split (into several configurations), merge (with others, into one). A configuration of two pilots that loses one ends the group.

Rates are configurations ending per hour spent at each size. Bands: 5–95% of 200 draws of the dependency blocks of the note, or of the dates where the captions say so.

2 What groups are

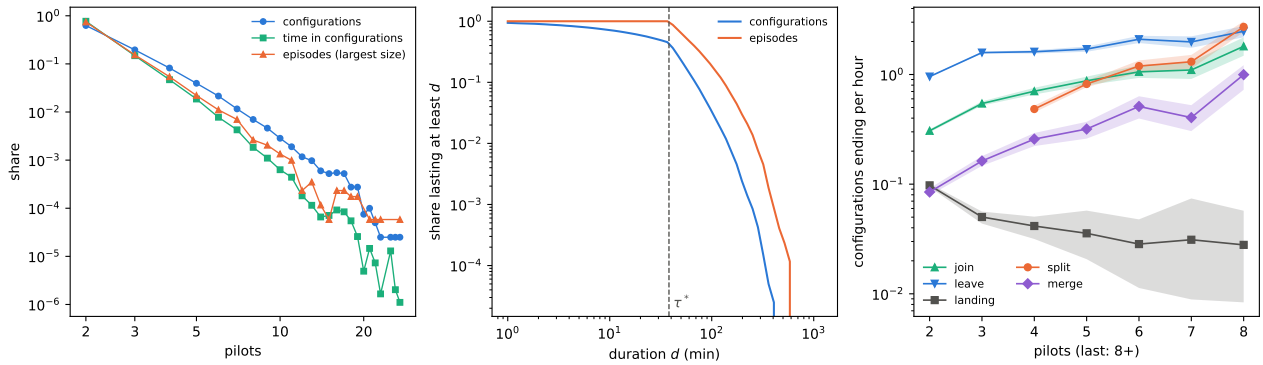


Figure 1: Left: share of configurations, of the time in configurations and of episodes (by their largest configuration), by number of pilots. Middle: share of configurations and of episodes lasting at least d; dashed: τ^*. Right: configurations ending per hour, by outcome and size (8+ pooled; a split needs 4 pilots), with 5–95% bands.

pilots	hours	median (min)	join	leave	landing	split	merge
2	17 389	41	0.31 [0.30, 0.32]	0.96 [0.95, 0.97]	0.098 [0.094, 0.10]	–	0.085 [0.079, 0.092]
3	3 349	20	0.54 [0.52, 0.57]	1.6 [1.6, 1.6]	0.050 [0.044, 0.056]	–	0.16 [0.15, 0.18]
4	1 059	13	0.70 [0.66, 0.76]	1.6 [1.6, 1.7]	0.042 [0.032, 0.051]	0.48 [0.45, 0.52]	0.26 [0.22, 0.29]
5	421	11	0.88 [0.80, 0.96]	1.7 [1.6, 1.8]	0.036 [0.021, 0.057]	0.82 [0.76, 0.88]	0.32 [0.26, 0.37]
6	176	8	1.1 [0.93, 1.2]	2.1 [1.9, 2.3]	0.028 [0.011, 0.048]	1.2 [1.0, 1.3]	0.51 [0.40, 0.63]
7	96	7	1.1 [0.91, 1.3]	2.0 [1.8, 2.2]	0.031 [0.0089, 0.074]	1.3 [1.1, 1.5]	0.40 [0.30, 0.53]
8+	107	4	1.8 [1.5, 2.2]	2.5 [2.2, 2.8]	0.028 [0.0084, 0.057]	2.7 [2.4, 3.1]	1.0 [0.73, 1.2]

Table 1: By size: hours spent, median duration of a configuration, and configurations ending per hour by outcome, with 5–95% bands (Figure 1, right).

Reading.

- **Most groups are pairs:** 63% of the 40 051 configurations and 77% of their time. The share falls steadily with size, up to 27 pilots (0.9% of the configurations hold a hang glider).
- **Most episodes never change:** 74% of the 17 169 episodes are one configuration; 5.0% hold a split, 4.4% a merge.
- **The episode lasts, the configuration does not.** An episode lasts 58 min (median), 12% of them 2 h or more; none is shorter than τ^*, by construction. A configuration lasts 41 min with 2 pilots and 4 min with 8 or more: joins, leaves, splits and merges all grow with size (Table 1), so a large configuration changes members every few minutes.
- **Groups break in flight:** 91% of the 18 376 configurations that end a group do so in flight, 9% at a landing. 34% of the 18 218 configurations with no predecessor form when a member departs: pilots who took off together.
- **A pair** ends 0.96 times per hour because the two separate, 0.31 because a third pilot joins, 0.085 by a merge and 0.098 by a landing.

2.1 Where in the flight groups form and break

On 1 Hz flights, where the phases are labelled: the phase at the instant a flight enters (31 381) or leaves (31 202) a group, against the shares of time in each phase; entries at the departure (1 179) and exits at the landing (1 358) are counted apart (Table 2).

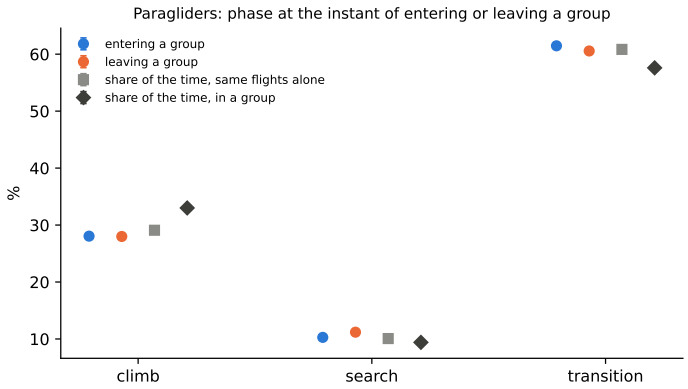


Figure 2: Phase at entry and exit, against the time shares (bands: blocks).

> **Reading.** Entries fall in climbs 28.1%, searches 10.3% and glides 61.5% of the cases, exits 28.0, 11.2, 60.6%, against 29.1, 10.1, 60.8% of the time alone: **no phase where groups preferentially form or break**. 34% of the entries are within 10 min of the departure.

2.2 When in the day

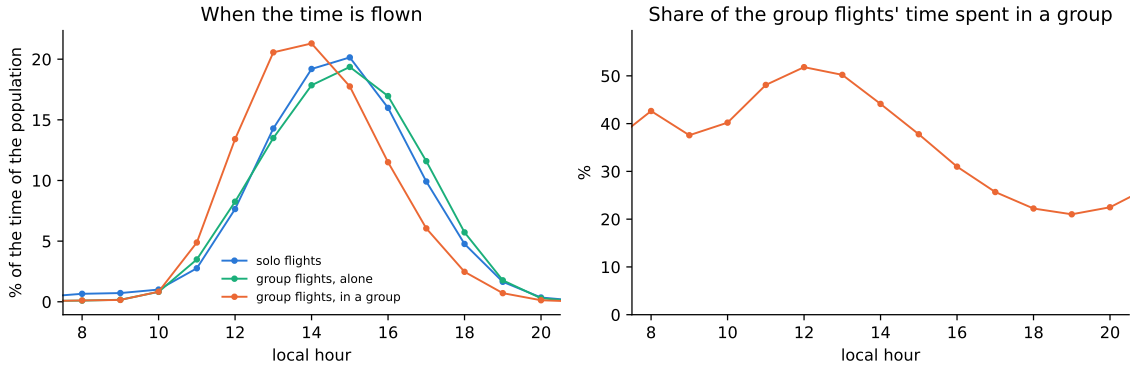


Figure 3: Time of solo flights (S), of group flights alone (A) and in a group (G) over the local hour (left), and the share of a group flight's time spent in a group (right).

> **Reading.** Group time is concentrated around midday: 52% of a group flight's time at 12 h, 38% at 9 h, 22% at 18 h.

3 Cohesion: what keeps two pilots together

Inside a contact, the delay $\ell^*(t)$ says how many seconds one pilot is behind the other on the same path (outside searches, where altitude does not fix it). Groups: the 34 082 pair intervals of pilots who took off within 10 km. Chance: the 134 202 contacts of the surrogate pairs of the null model (the same pairs as in the note), two pilots who never saw each other; the 980 that last at least τ^* are selected as the groups are, by staying in contact that long.

- $D(s) = \langle(\ell(t+s) - \ell(t))^2\rangle$, the mean square change of the delay after s, fitted by an Ornstein–Uhlenbeck process (Appendix B), $D(s) = 2\sigma^2(1 - e^{-ks}) + c$: $1/k$ is how long the delay remembers its value, σ its spread, $2k\sigma^2$ how fast it wanders.
- The drift $\langle\ell(t + 60\text{ s}) - \ell(t)\rangle/60$ s at each ℓ, made symmetric since the order of a pair is arbitrary: the pull back to $\ell = 0$, a spring $-k\ell$ near 0.
- The distance between the two pilots at the same instant after their first contact.

Two independent pilots separate as twice the displacement variance of one, not twice its MSD: they share the drift along the route. The surrogate pairs give this baseline directly, with the same terrain.

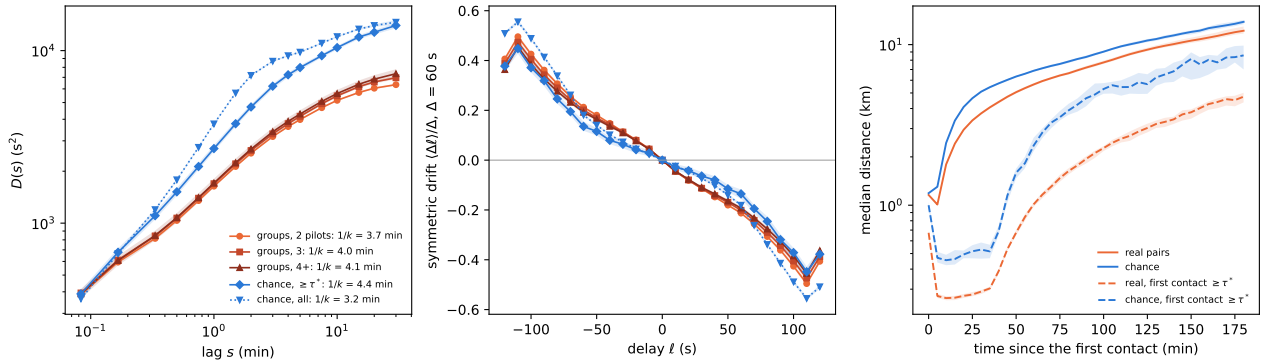


Figure 4: Left: $D(s)$ of groups (by size) and of chance contacts, with $1/k$ of the Ornstein–Uhlenbeck fit. Middle: symmetric drift of the delay. Right: median distance at the same instant after the first contact, for every pair and for those whose first contact lasts at least τ^*. Bands: 5–95% of 200 draws of the dates.

Reading.

- **Groups keep the same pace.** Their delay wanders at $2k\sigma^2 = 24.5$ [23.9, 25.1] s^2/s, against 46 [44, 48] for chance contacts as long and 69 [68, 70] for all of them; its spread is $\sigma = 54$ s [53, 54], against 78 s [77, 79].
- **They are not pulled back much harder.** The delay forgets itself in $1/k = 3.9$ min [3.8, 4.0], against 4.4 min [4.2, 4.7] by chance; the drift gives 4.7 min [4.6, 4.7] against 6.4 min [5.9, 6.9]. Most of the pull is the selection: a contact that lasts keeps $|\ell| \leq \delta$ by definition.
- **Size does not matter:** $2k\sigma^2 = 24.2$, 24.5 and 25.0 s^2/s at 2, 3 and 4+ pilots.
- **They stay together after the contact that selected them.** When the first contact lasts at least τ^*, the two pilots are 1.0 km apart 60 min later (median), against 2.4 km by chance; 2.9 against 5.8 km after 2 h. Over every first contact, 3.7 against 5.1 km after 30 min.

A group is two pilots who keep the same pace, not two pilots tied by a spring. Whether this is following or the same choices in the same air, the data cannot tell.

4 Shape: chains or compact groups

Inside a configuration the members are fixed, but the pair intervals among them start and end. Between two such changes the graph has a *density* (the share of the member pairs in contact) and a *diameter* (the most steps between two members); its shape is *complete* (every member in contact with every other), a *tree* (the fewest edges that keep the members connected: a chain or a star) or between (Appendix A). The TA-MSD pieces of *Group and solo flights* lie inside one configuration and take the shape of the graph at their middle.

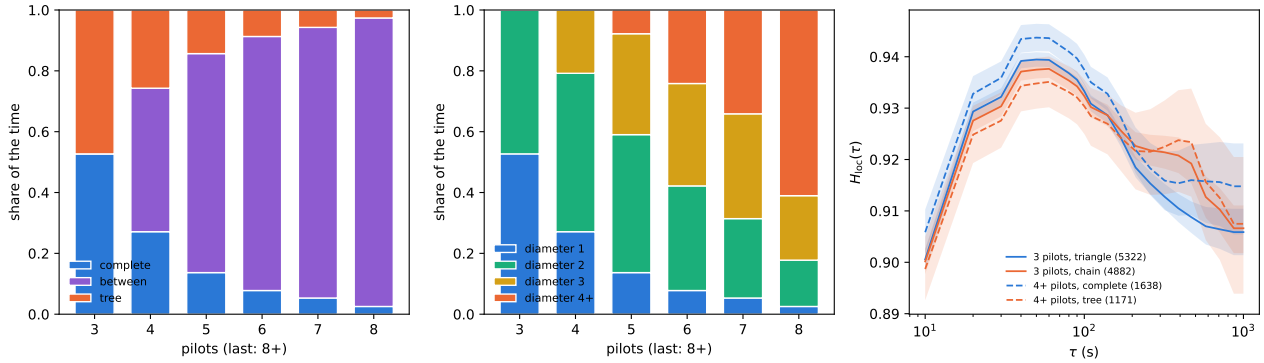


Figure 5: Left: share of the time of each shape, by size (8+ pooled). Middle: share of the time by diameter. Right: $H_{\mathrm{loc}}(\tau)$ of paragliders in triangles and chains of 3 pilots, and in complete and tree graphs of 4 or more (flights in brackets), with 5–95% bands.

Reading.

- **Large groups are strings, not clouds.** Three pilots form a triangle 53% of the time and a chain 47%. Complete graphs fall from 27% of the time at 4 pilots to 2.5% at 8+, the mean density from 0.84 at 3 to 0.45 at 8+; at 8+ the diameter is 4 or more 61% of the time: pilots follow each other along the route.
- **The shape does not change the flight.** $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.906 [0.901, 0.910] in triangles and 0.907 [0.902, 0.911] in chains; 0.915 [0.905, 0.923] in complete graphs of 4+ and 0.907 [0.894, 0.920] in trees. The straighter progress of *Group and solo flights* does not need every pilot in contact with every other.

5 Who leads

The signed median delay of a pair interval says who passes first: the pilot ahead, when the other passes at least 15 s later; otherwise the two fly abreast (35% of the pairs of two; median delay 20 s). How leading changes the flight is in *Group and solo flights*. Here:

- **Order** in configurations of 3 pilots or more: each edge points from the pilot ahead to the one behind, a directed graph (Appendix A), against the same graphs with each direction drawn at random (50 draws);
- **switches** of the pilot ahead along $\ell^*(t)$ in the pair intervals of Section 3 (stretches shorter than 60 s dropped), against chance contacts lasting at least τ^*;
- **who leads** in configurations of two named pilots: earlier flights in the catalog and wing class (A = 1 to competition = 5);
- **habitual leaders:** for the 946 pilots with at least 10 roles in pairs, the dispersion of their lead fractions, against roles swapped at random.

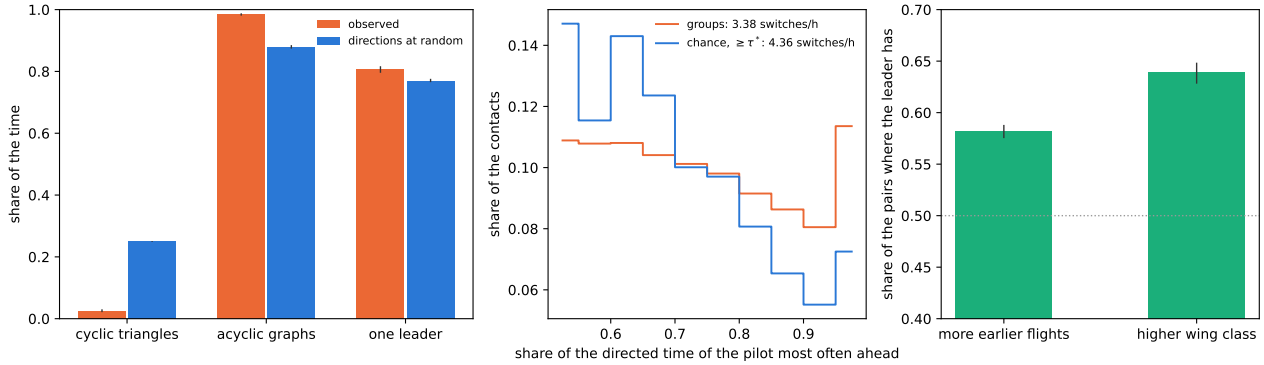


Figure 6: Left: in groups of 3 or more, share of the time with cyclic directed triangles, acyclic graphs (every edge directed) and one leader (acyclic graphs), observed and with directions at random. Middle: share of the directed time of the pilot most often ahead in each contact, with the switches per hour. Right: share of the pairs of two in which the leader has more earlier flights, or the higher wing class. Bars: 5–95% (blocks of the note).

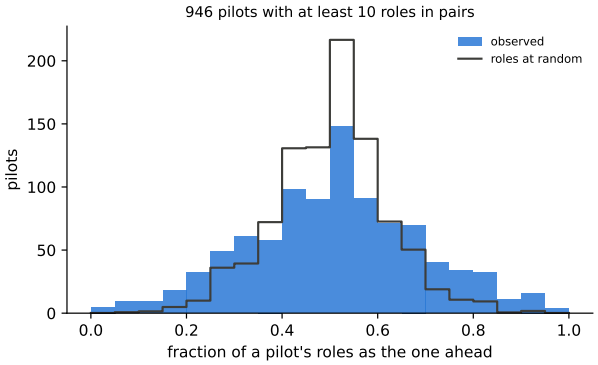


Figure 7: Lead fraction of each pilot, against roles at random.

Reading.

- **Pilots are in a line.** Directed triangles are cyclic 2.5% [2.1, 3.0] of the time, against 25% at random; graphs with every edge directed (33% of the time) are acyclic 98.5%, against 88%. Positions along one route cannot form cycles: this is geometry, not rank.
- **One leader, in small groups:** 80.6% [79.6, 81.7] of the acyclic time, against 77.0% [76.5, 77.6] at random; 86% at 3 pilots, 68% at 4, 50% at 5 or more. A single file (depth $n-1$): 71, 46 and 25%.
- **The lead changes often.** The pilot ahead changes 3.4 times per hour of contact [3.3, 3.4], against 4.4 [4.2, 4.5] by chance; the pilot most often ahead holds 73% of the directed time (median), against 69%.
- **The experienced pilot on the faster wing leads.** The leader has more earlier flights in 58.2% [57.5, 58.8] of 15 937 pairs (1.3 times as many, median), the higher wing class in 63.9% [62.8, 64.9] of 8 304.
- **Some pilots lead or follow persistently:** dispersion of the lead fractions 2.73 against 1.00 at random (95% below 1.07); 12% lead in at most 20% or at least 80% of their roles, against 3% expected.

6 Repeated partners

Paraglider pilots by their name in the catalog (anonymised ones left out). Two pilots are partners on a day when their flights share a pair interval: a *pair-day*. Chance keeps who flies in a group where and when: within each site-day (takeoffs within 3 km, joined when a pair interval links two sites), the partners are redrawn at random, each pilot keeping its number of partners that day (a configuration model, Appendix A; 100 draws).

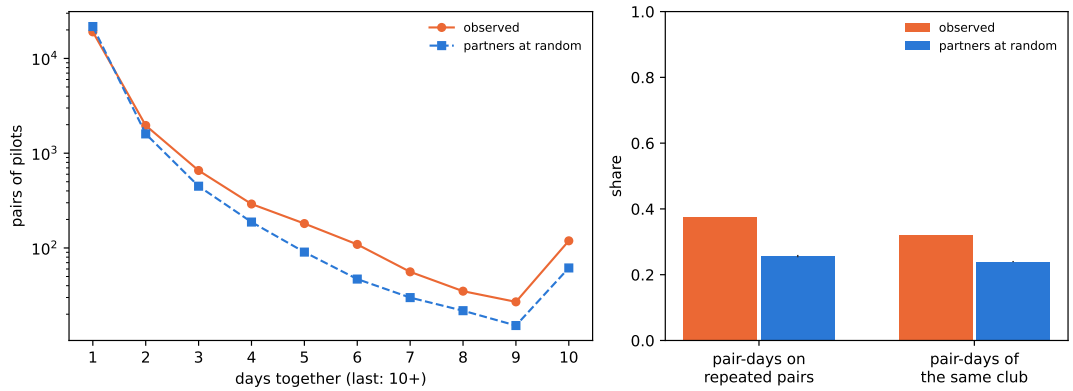


Figure 8: Left: pairs of pilots by the number of days they flew together, observed and with partners at random. Right: share of the pair-days on pairs that meet on more than one day, and of two pilots of the same club (each pilot's main club); bars at random: 5–95% of the draws.

Reading. 30 550 pair-days among 4 525 pilots and 22 513 pairs; one pair flew together on 69 days.

- **Pilots return to the same partners:** 37.6% of the pair-days are on pairs that meet on more than one day, against 25.7% [25.3, 26.0] at random (3 437 such pairs against 2 501).
- **and to their club:** 32.1% of the pair-days join two pilots of the same club, against 24.0% [23.8, 24.2].

Chance already fixes who flies in a group at each site and day, so the excess is the choice of partner among the pilots there. Choosing where and when to fly with friends is not removed: it is part of chance here.

7 Fission and fusion: where the sizes come from

$T(n)$ is the share of the time spent in configurations of n pilots: the chance that a group seen at a random instant has n pilots (Figure 1, left). If sizes change one pilot at a time, the flux from n to $n+1$ pilots balances the flux back, a birth–death chain (Appendix B):

$$\frac{T(n+1)}{T(n)} = \frac{g(n)}{l(n+1)}, \tag{1}$$

with $g(n)$ the joins and $l(n)$ the leaves and landings per hour at size n (Table 1). Splits and merges are left out. Per member, $g(n) = a\,n$ and $l(n) = \mu\,n^{1-\gamma}$: every member brings in others at rate a and leaves at rate $\mu n^{-\gamma}$, fitted to the rates, not to T. With $\gamma = 0$ the chain gives the log-series $T(n) \propto x^n/n$, $x = a/\mu$. For comparison, the form of [1], $T(n) \propto n^{-\alpha} e^{-n/n_c}$, fitted to T.

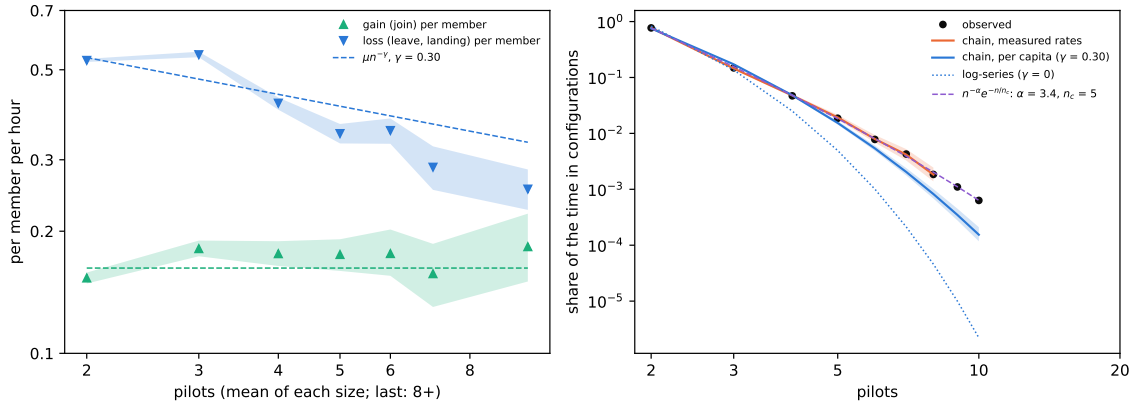


Figure 9: Left: joins and losses per member per hour by size (mean size of each class; 8+ pooled), with a and $\mu n^{-\gamma}$ dashed. Right: $T(n)$ observed (sizes with at least 10 h) and from the chain (1) with the measured rates, with the per-capita rates and with $\gamma = 0$; dashed purple: the form of Bonabeau et al. fitted to T. Bands: 5–95%.

Reading.

- **One pilot at a time is enough.** With the measured rates the chain gives $T(n)$ within 4% from 2 to 8 pilots (0.145 against 0.148 at 3 pilots, just outside its band), although splits and merges are 6% of the changes at 2 pilots and 46% at 8+.
- **Joining grows with the members, leaving does not.** Each member brings in others at $a = 0.162$ [0.158, 0.167] per hour, at every size. Each leaves at 0.53 per hour in a pair and 0.25 at 8+: $\gamma = 0.30$ [0.28, 0.33].
- **The heavy tail comes from staying.** With $\gamma = 0$ ($x = 0.25$) $T(10)$ is 290 times too small; with $\gamma = 0.30$, 4 times. Fitted directly, $\alpha = 3.4$ and $n_c = 5$.

8 Conclusions and limits

Supported

- groups are mostly pairs; large ones are strings along the route that change members every few minutes, while the episode lasts about an hour;
- group pilots keep the same pace: their delay wanders half as fast as that of two pilots kept in contact by chance, and they stay closer after the contact;
- the size distribution follows from one pilot joining or leaving at a time; its heavy tail from members of larger groups leaving less often;
- the lead changes a few times per hour and goes more often to the experienced pilot on the higher wing;
- pilots return to the same partners and to their club mates more than who flies where and when explains.

Not supported

- a spring that pulls pilots back: the pull is about that of chance contacts selected the same way;
- that the shape of a group (triangle or chain, complete or tree) changes how its pilots fly;
- a social rank from the order in a group: a line along the route is geometry;
- that keeping pace means following: the surrogate pairs fly at another hour, so pilots in the same air at the same time may share the pace without seeing each other;
- pilots are names of the catalog: homonyms join, a pilot who changes name splits.

References

[1] E. Bonabeau, L. Dagorn, P. Fréon, Scaling in animal group-size distributions, *PNAS* 96, 4472–4477 (1999).

[2] A. Flack, M. Nagy, W. Fiedler, I. D. Couzin, M. Wikelski, From local collective behavior to global migratory patterns in white storks, *Science* 360, 911–914 (2018).

[3] M. Nagy, Z. Ákos, D. Biro, T. Vicsek, Hierarchical group dynamics in pigeon flocks, *Nature* 464, 890–893 (2010).

[4] M. E. J. Newman, *Networks: An Introduction*, Oxford University Press (2010).

[5] M. Niu, P. G. Blackwell, A. Skarin, Modeling interdependent animal movement in continuous time, *Biometrics* 72, 315–324 (2016).

[6] M. Niu, F. Frost, J. E. Milner, A. Skarin, P. G. Blackwell, Modelling group movement with behaviour switching in continuous time, *Biometrics* 78, 286–299 (2022).

[7] G. E. Uhlenbeck, L. S. Ornstein, On the theory of the Brownian motion, *Phys. Rev.* 36, 823–841 (1930).

A Graphs, from scratch

- **Graph.** A set of *nodes* and a set of *edges*, each edge joining two nodes. Here, at each instant, a node is a flight and an edge a pair interval in progress.
- **Path.** A sequence of nodes, each joined by an edge to the next. Two nodes are *connected* when a path joins them.
- **Connected component.** A largest set of nodes connected to each other. Every node is in exactly one component; a node with no edge is a component alone (a pilot flying solo). A *configuration* is a component with at least two nodes.

Two members of a configuration need not share an edge: in the chain 1–2–3 of Figure 10, pilots 1 and 3 are in the same configuration through pilot 2.

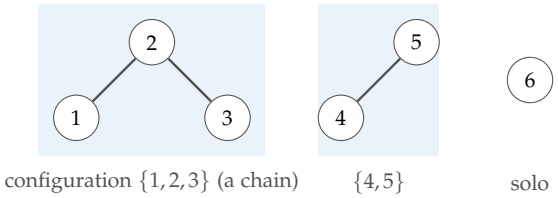


Figure 10: The graph at one instant: three components, two of them configurations.

- **Directed graph.** The edges carry an arrow. For the episodes the nodes are configurations, with an arrow $c \to c'$ when c' starts as c ends and they share a member. An episode is a connected component of this graph, arrows ignored.
- **Bipartite graph.** Nodes of two kinds, edges only between kinds. At one instant, the configurations ending and those starting, with an edge when they share a member. Each component is one event (Figure 11); its outcome follows from how many configurations it has on each side: one to one is a join or a leave (members added or removed), one to several a split, several to one a merge.

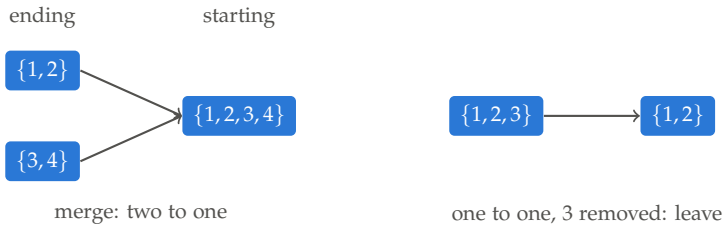


Figure 11: Two events: each is a connected component of the bipartite graph of the configurations ending and starting at one instant.

Shape (Section 4). For a connected graph of n nodes and m edges:

- **Density** $m/\binom{n}{2}$: the share of the node pairs joined by an edge.
- **Distance** between two nodes: the fewest edges of a path joining them. **Diameter**: the largest distance. In the chain 1–2–3 the diameter is 2; in a triangle, 1.
- **Complete graph**: every pair joined, density 1, diameter 1. **Tree**: $m = n - 1$, the fewest edges that keep n nodes connected, with no closed path; a *chain* is a tree with diameter $n - 1$, a *star* one with diameter 2.

Order (Section 5). In a directed graph:

- a **cycle** is a path that follows the arrows back to its start (A→B→C→A); a graph with none is **acyclic**, and its nodes can be put in a line with every arrow pointing forward;
- a **source** has no arrow coming in: here, a pilot with nobody ahead, a leader;
- the **depth** of an acyclic graph is its longest chain of arrows: 1 for a star, $n - 1$ for a line.

A directed triangle is *transitive* (A→B, B→C, A→C) or *cyclic*; with arrows drawn at random, 2 of its 8 orientations are cyclic.

Configuration model (Section 6). A random graph with the *degrees* (the number of edges of each node) of a given one: each node gets as many half-edges as its degree, and the half-edges are paired at random [4]. It keeps how many partners each pilot had and changes only who they were.

B Two models

Ornstein–Uhlenbeck process [7] (Section 3). A quantity ℓ pulled back to 0 by a spring of rate k and shaken by noise:

$$d\ell = -k\,\ell\,dt + \sqrt{2k}\,\sigma\,dW. \tag{2}$$

It forgets its value after $1/k$ and spreads by σ around 0. Its mean square change after a lag s is $D(s) = \langle(\ell(t+s) - \ell(t))^2\rangle = 2\sigma^2(1 - e^{-ks})$: it grows as $2k\sigma^2 s$ at short lags, like a free random walk, and stops at $2\sigma^2$ after $1/k$. With $k = 0$ it is a random walk and D grows forever. A measurement noise of variance $c/2$ on each value adds c to $D(s)$ at every lag.

Birth–death chain (Section 7). A group of n pilots gains one at rate $g(n)$ and loses one at rate $l(n)$. Over a long time the transitions from n to $n+1$ and back are equally many, so the time spent at $n+1$ and at n, $T(n+1)$ and $T(n)$, satisfy $T(n)\,g(n) = T(n+1)\,l(n+1)$, Eq. (1). Multiplying the ratios from 2 to n gives $T(n)$. If every member joins others at rate a and leaves at rate μ, $g(n) = an$, $l(n) = \mu n$ and $T(n) \propto x^n/n$ with $x = a/\mu$, the log-series: the tail falls exponentially. A leaving rate that falls with n makes it heavier.

C Phase at entry and exit

	climb (%)	search (%)	transition (%)
entering a group	28.1 [27.6, 28.5]	10.3 [10.0, 10.6]	61.5 [61.0, 61.9]
leaving a group	28.0 [27.6, 28.3]	11.2 [10.9, 11.5]	60.6 [60.1, 61.0]
share of time, alone	29.1 [28.9, 29.3]	10.1 [10.0, 10.2]	60.8 [60.6, 61.0]
share of time, in a group	33.0 [32.8, 33.2]	9.4 [9.2, 9.6]	57.6 [57.3, 57.9]

Table 2: Phase at the instant of entering or leaving a group, and shares of time in each phase (percent, with 5–95% bands; 1 Hz flights).

D Reproducing the report

From `analyses/groups`, after the scripts of *Group and solo flights*:

```
uv run python describe.py     # sizes, durations, events; about 15 s
uv run python cohesion.py     # delays and separations, one pass over every date; about 25 min
uv run python chains.py       # shapes and their H_loc; about 40 s
uv run python hierarchy.py    # order, switches, who leads; after cohesion.py
uv run python partners.py     # repeated partners; about 4 min
uv run python fission.py      # the birth-death chain; about 10 s
uv run python report/tables.py; uv run python report/structure_tables.py
cd report && latexmk -pdf group_structure.tex
```

The per-date cache of `cohesion.py` is in `analyses/cohesion/all/days/` of the run on SSD_DISANTE.